

AS level Economics

Chp:- Demand and Supply

Topic:- Indirect tax and PED!

alpphp

Zarak
Mustafay



Indirect tax

e.g. sales tax /
Value added tax /
Duty / Tariff

Imposed on goods and
services

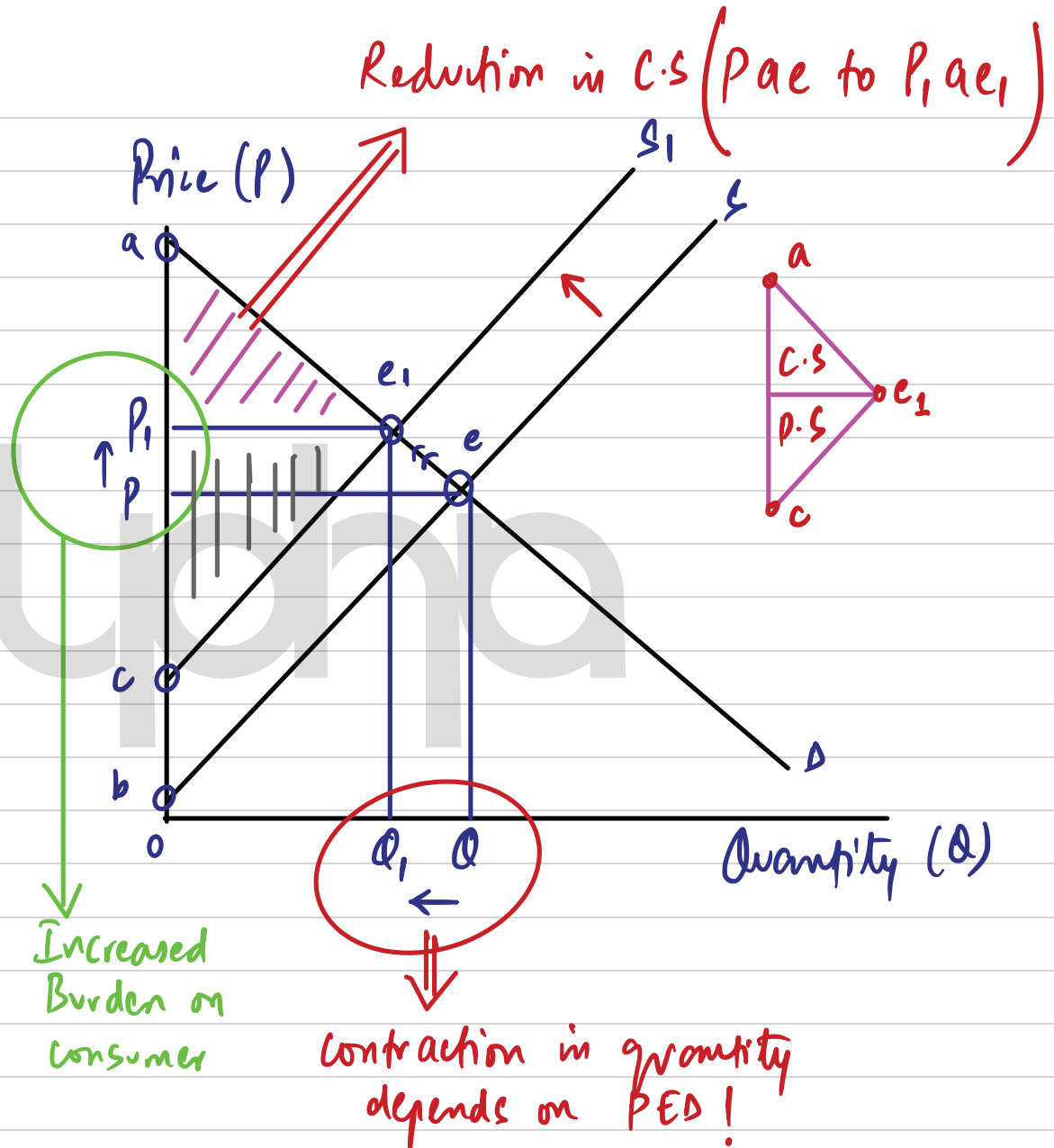
Cost of production ↑

Supply ↓ (Step #1)

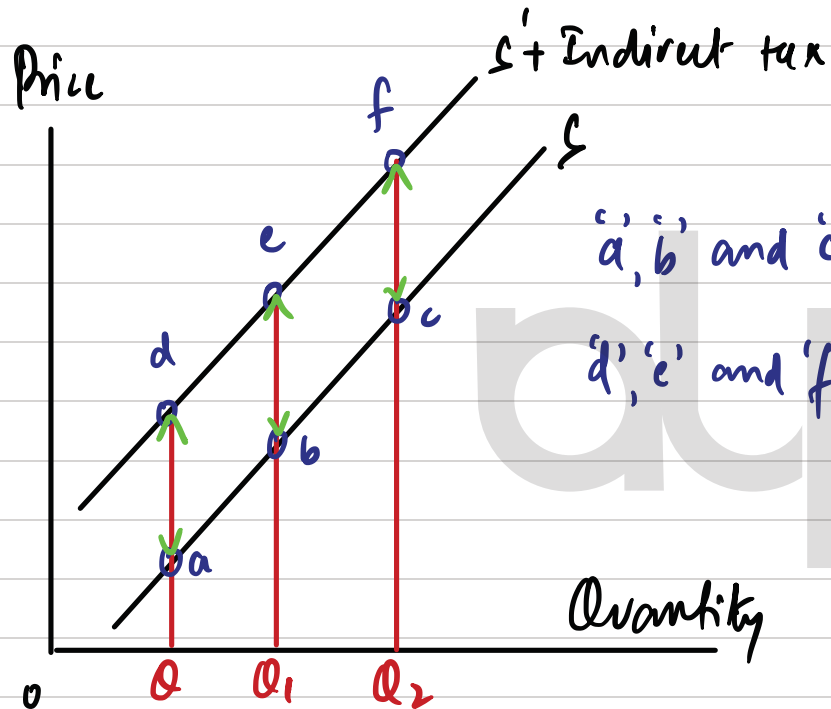
Burden on producer is
passed on to consumers
in form of higher prices

Quantity demanded ↓

To what extent:- Depends on PED!



Note:- It's important to note that when indirect tax is imposed, supply curve shifts parallelly leftwards, represents the vertical distance is same!

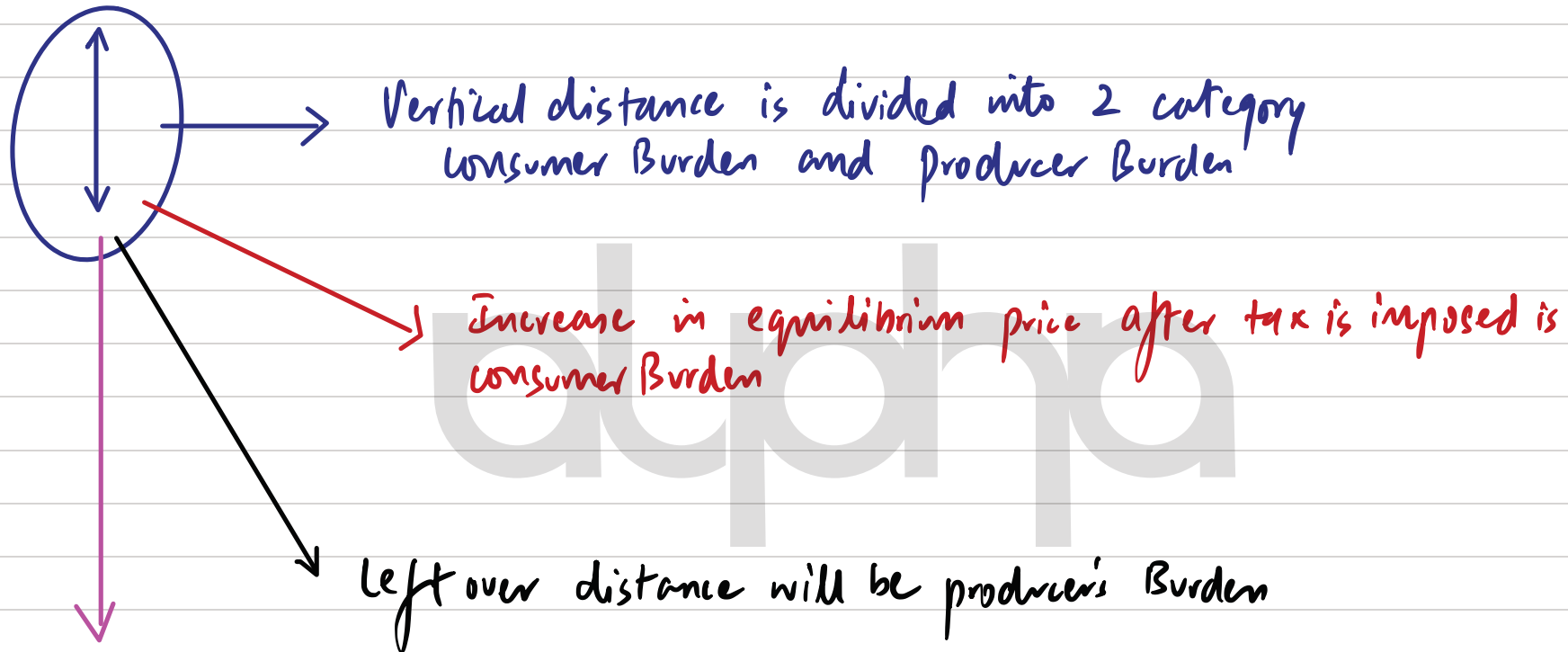


'd', 'e' and 'f' after tax!

'a', 'b' and 'c' before tax!

Thus the vertical distance between 2 supply curves is per unit tax / specific tax!

per unit tax | Specific tax



The extent to which burden will be shared depends upon price elasticity of demand.

Example to understand the vertical distance

- ★ C.B:- Consumer Burden
- ★ P.B:- Producer Burden

★ Video Tape, selling price = \$8.00

★ Indirect tax is imposed on every video tape = \$2.00

★ selling price after tax = ?

★ Most students will say \$10.00, but is it correct?

★ Producer sets the price after tax, S.P = \$8.50

Only \$0.50 was increased, even though, producer needs to pay \$2.00 to the government, why?

★ Can you involve PED?

★ Because the demand is elastic, availability of substitutes, the producer can't afford to pass on majority of the burden.

★ \$8.00 → \$8.50 → Consumer pay \$0.50, Government requires \$2.00
 Producer pay \$1.50 ⇒ P.B > C.B

Video Tape price after tax! Initial price \$8.00

1 \$8.50 \$2.00 C.B. :- \$0.50 (\$8.50 - \$8.00)
 P.B > C.B (PED > 1, Elastic)
 P.B :- \$1.50

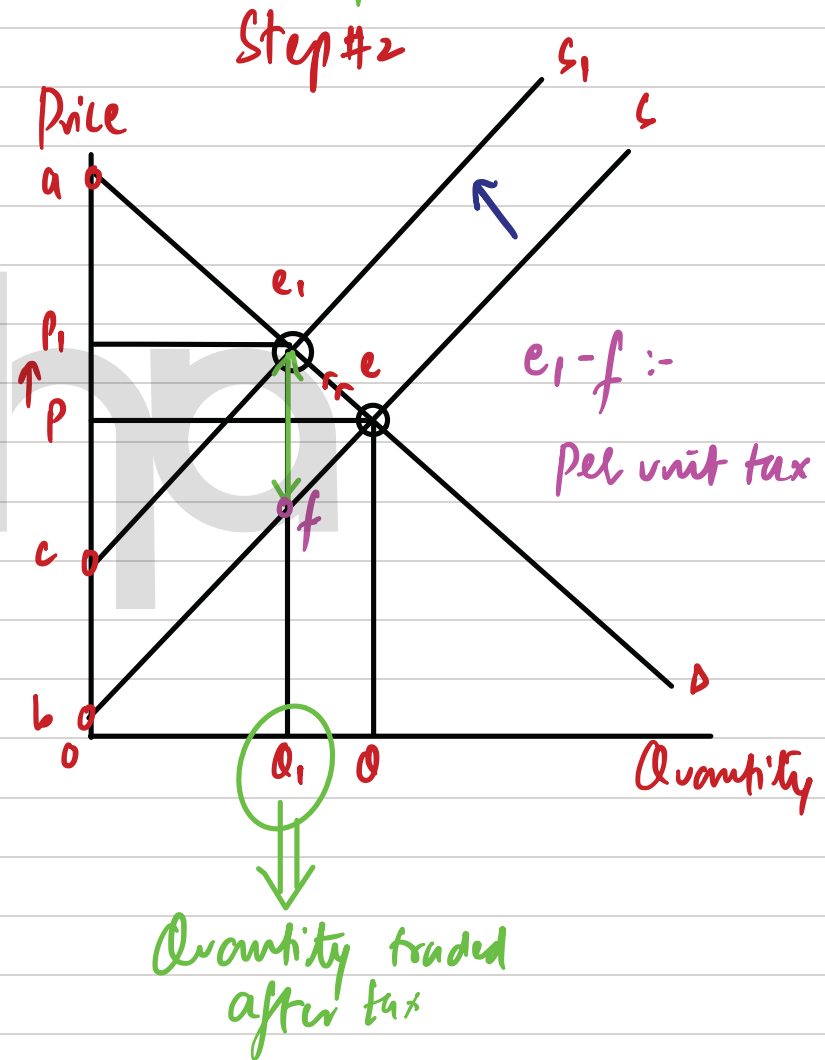
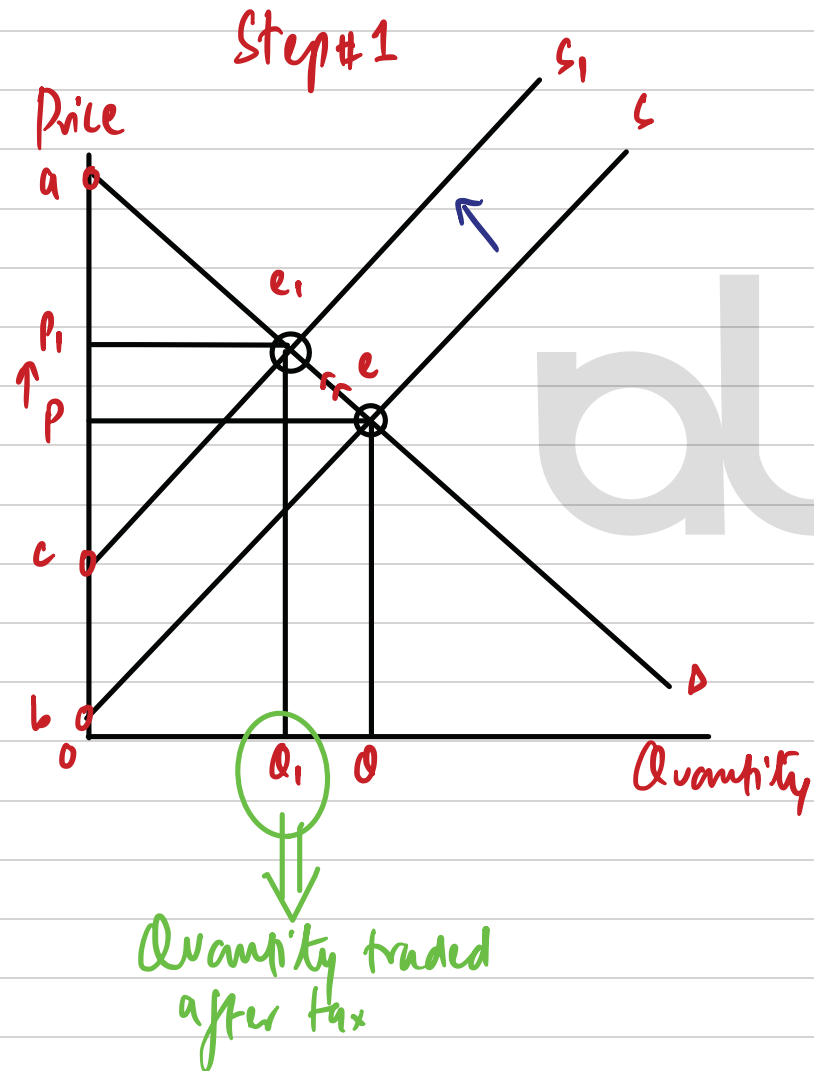
2 \$9.50 \$2.00 C.B :- \$1.50 (\$9.50 - \$8.00)
 C.B > P.B (PED < 1, Inelastic)
 P.B :- \$0.50

3 \$10.00 \$2.00 C.B :- \$2.00 (\$10.00 - \$8.00)
 Entire Burden on consumers (Perfectly inelastic)
 P.B :- \$0

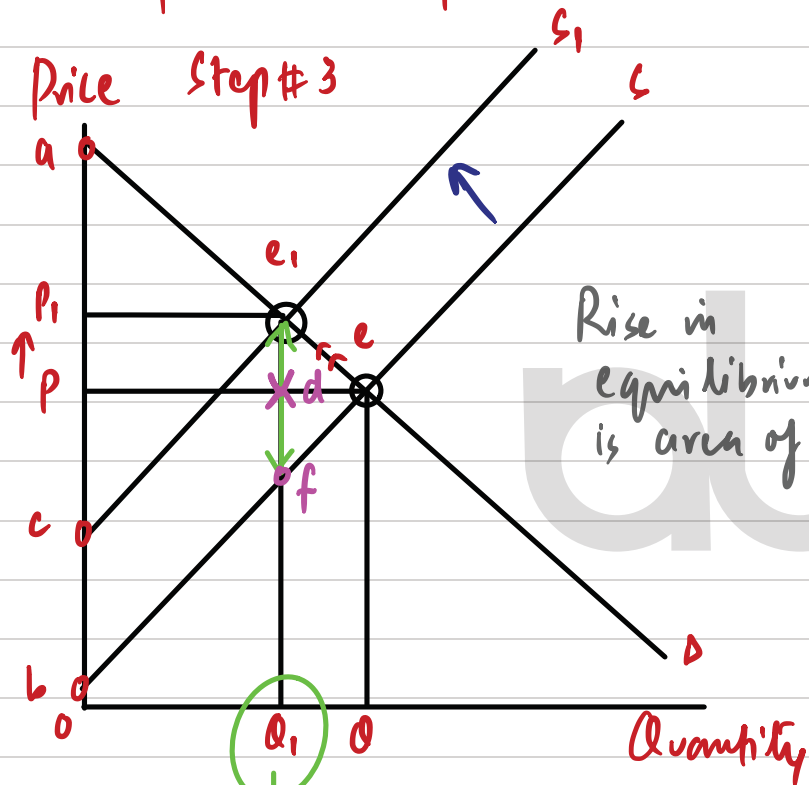
4 \$8.00 \$2.00 C.B :- \$0 (\$8.00 - \$8.00)
 Entire Burden on producer (perfectly elastic)
 P.B :- \$2.00

Intro diagram of Indirect tax

Identify Vertical distance!



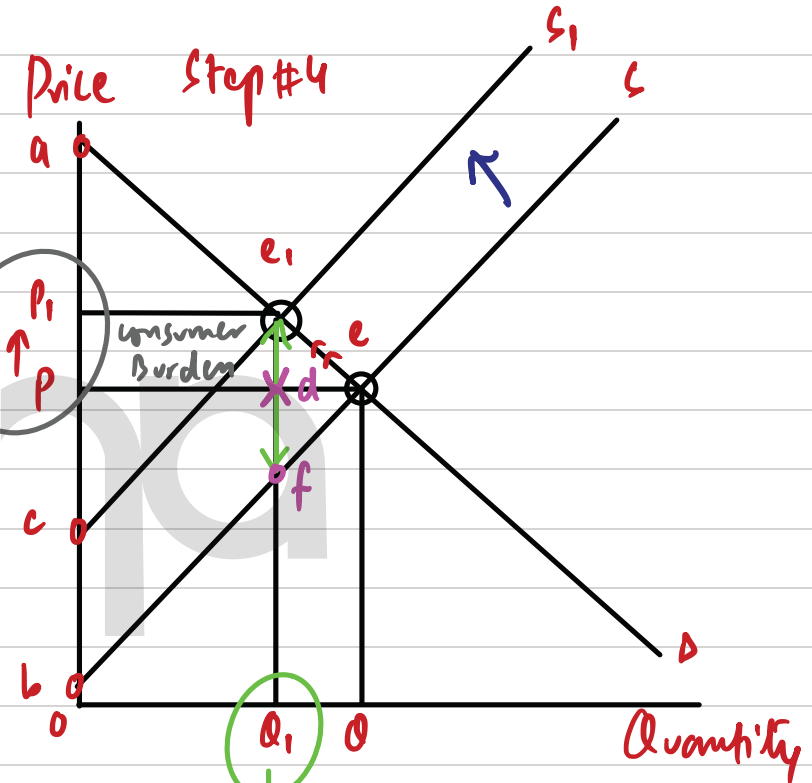
Divide the distance into 2,
from initial price 'P'



Quantity traded
after tax

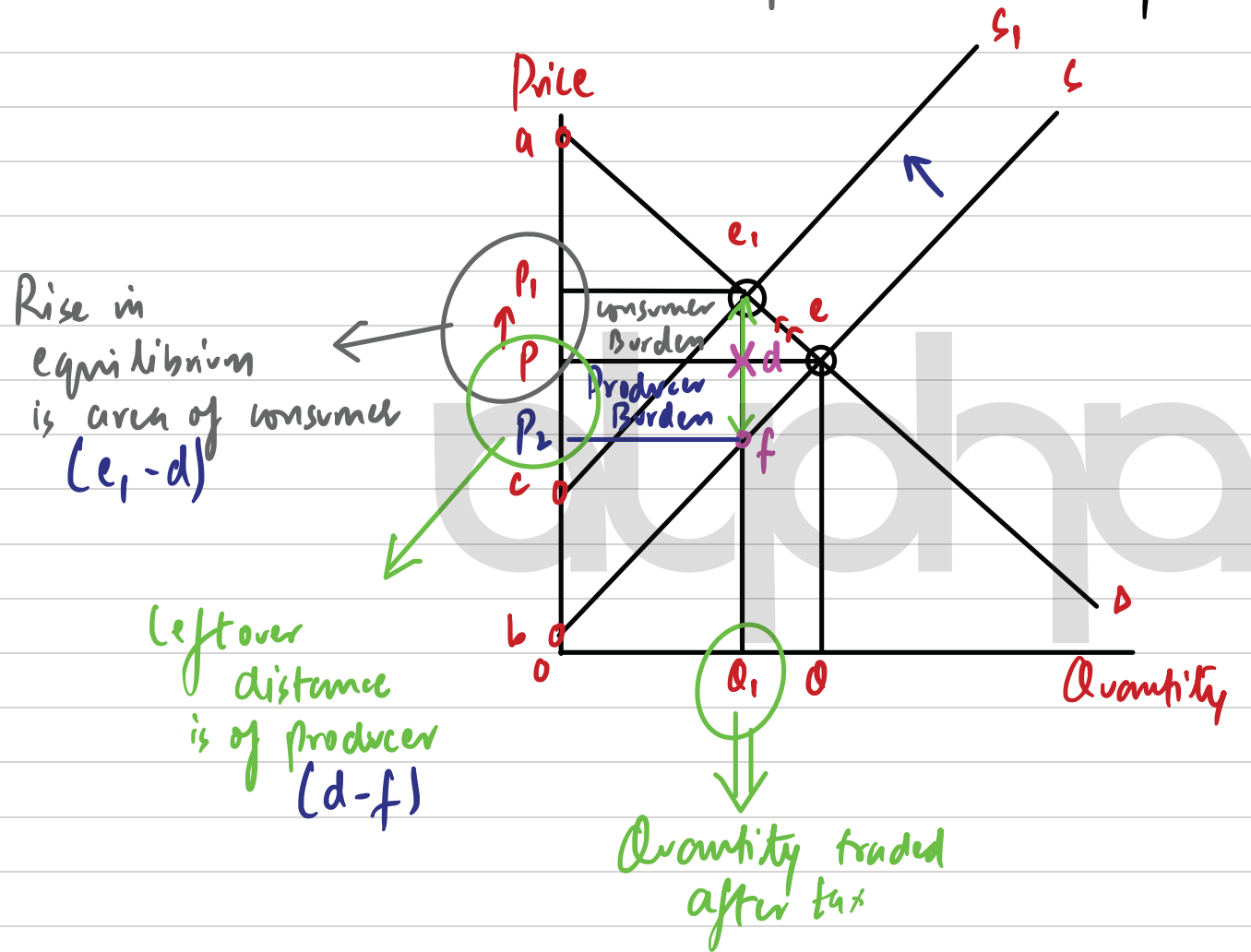
' $e_1 - d$ ' and ' $d - f$ '

Rise in
equilibrium
is area of consumer



Quantity traded
after tax

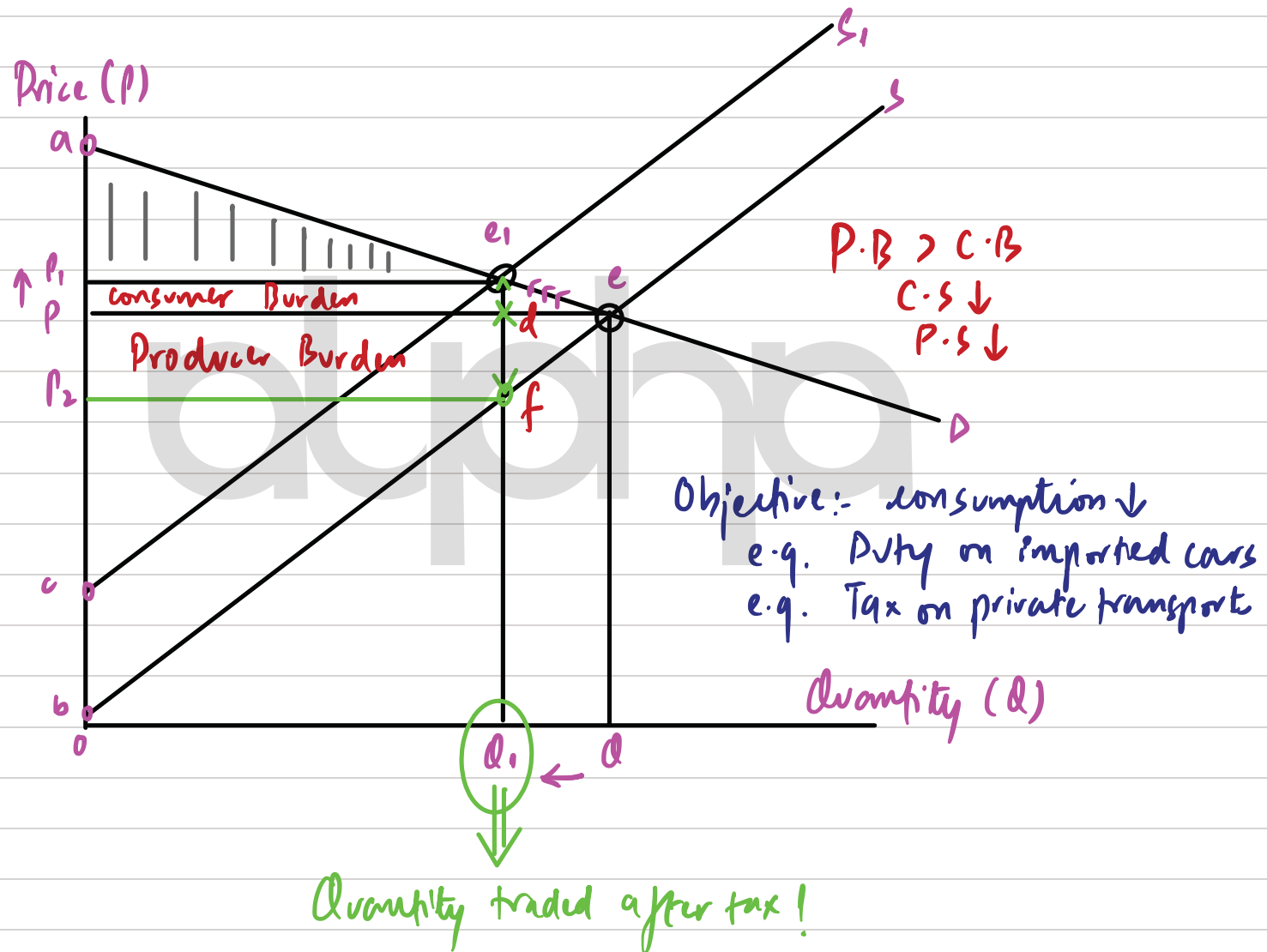
Step #5 (Main diagram)



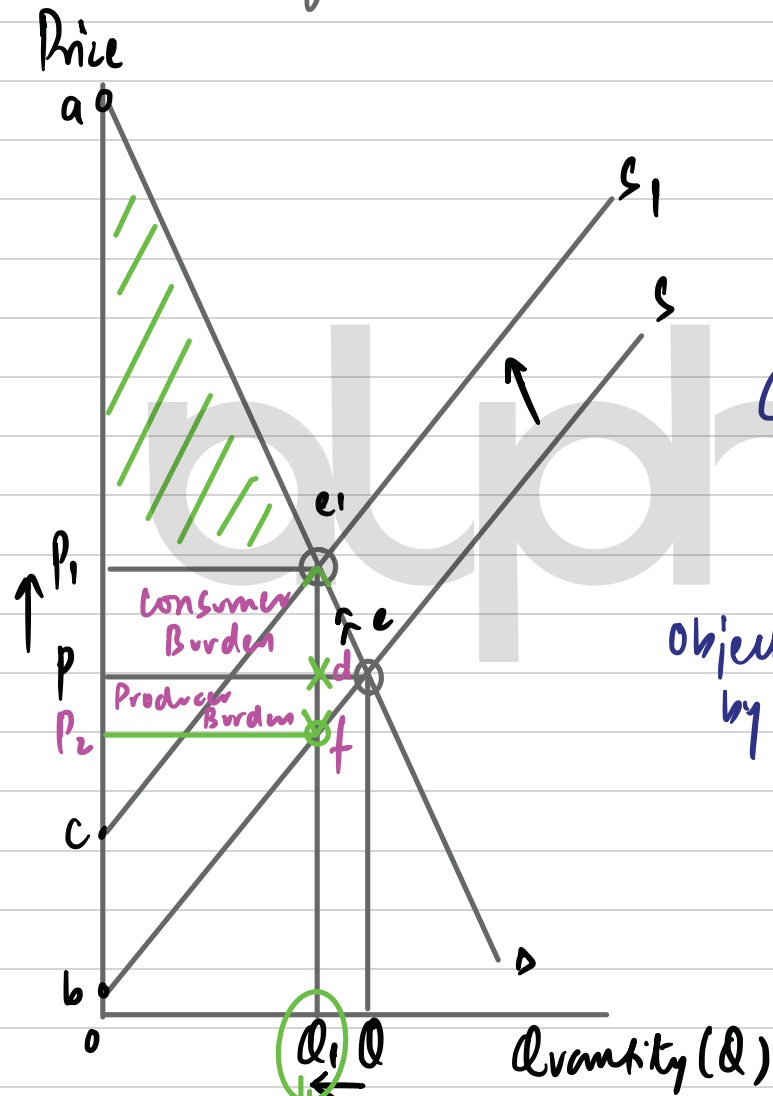
Pointers

- o Indirect tax imposed, Supply shifts left
- o Price $\uparrow$ Quantity traded $\downarrow$
- o Identify vertical distance on new quantity traded
- o Rise in equilibrium price, Consumer Burden
- o leftover distance, producer Burden
- o Impact on consumer and producer surplus, both reduced
- o Indirect taxes are regressive, impacts poor, distorts inequality and create inflationary pressure
- o Helps in generating tax revenue from harmful goods, which can be used in advertising to correct information failure

Indirect tax is imposed, Demand is elastic, $PE_D > 1$



Indirect tax imposed, if demand is inelastic, $PED < 1$



$C.B > P.B$

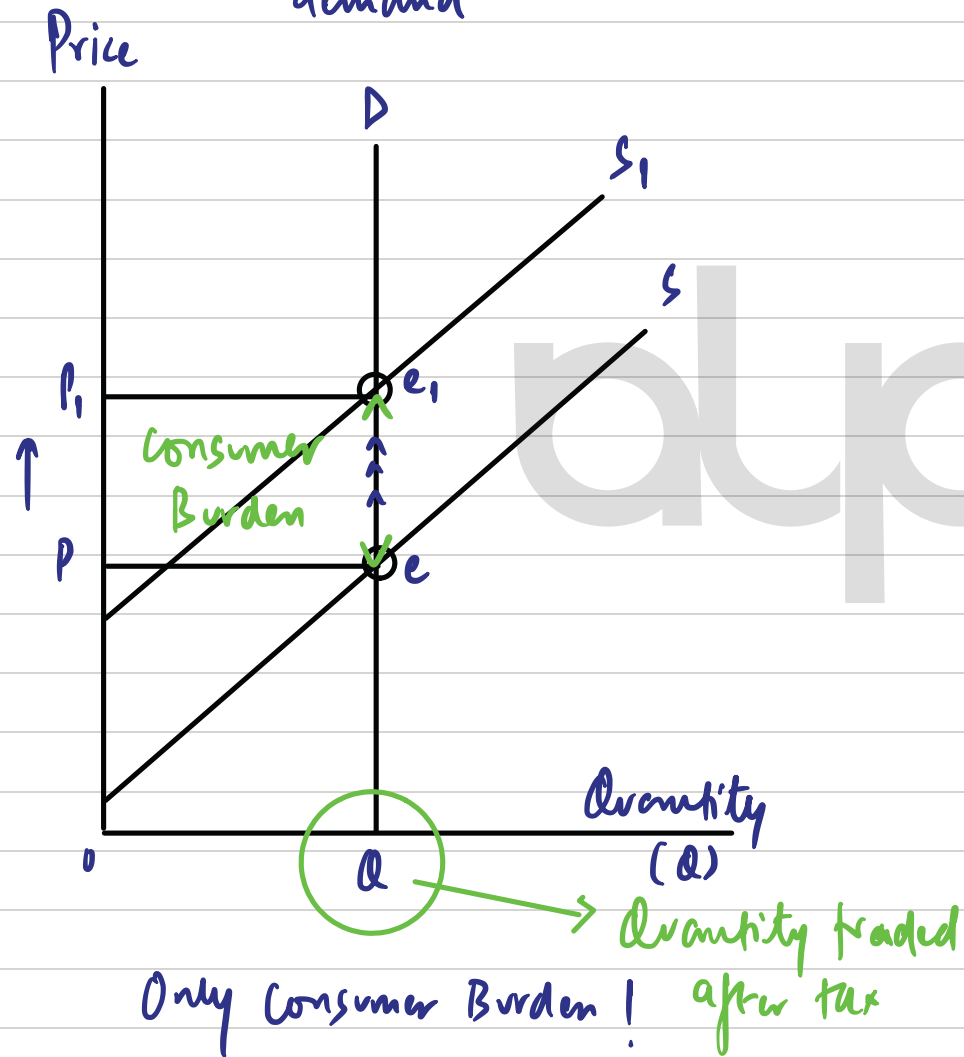
$C.S \downarrow$

$P.S \downarrow$

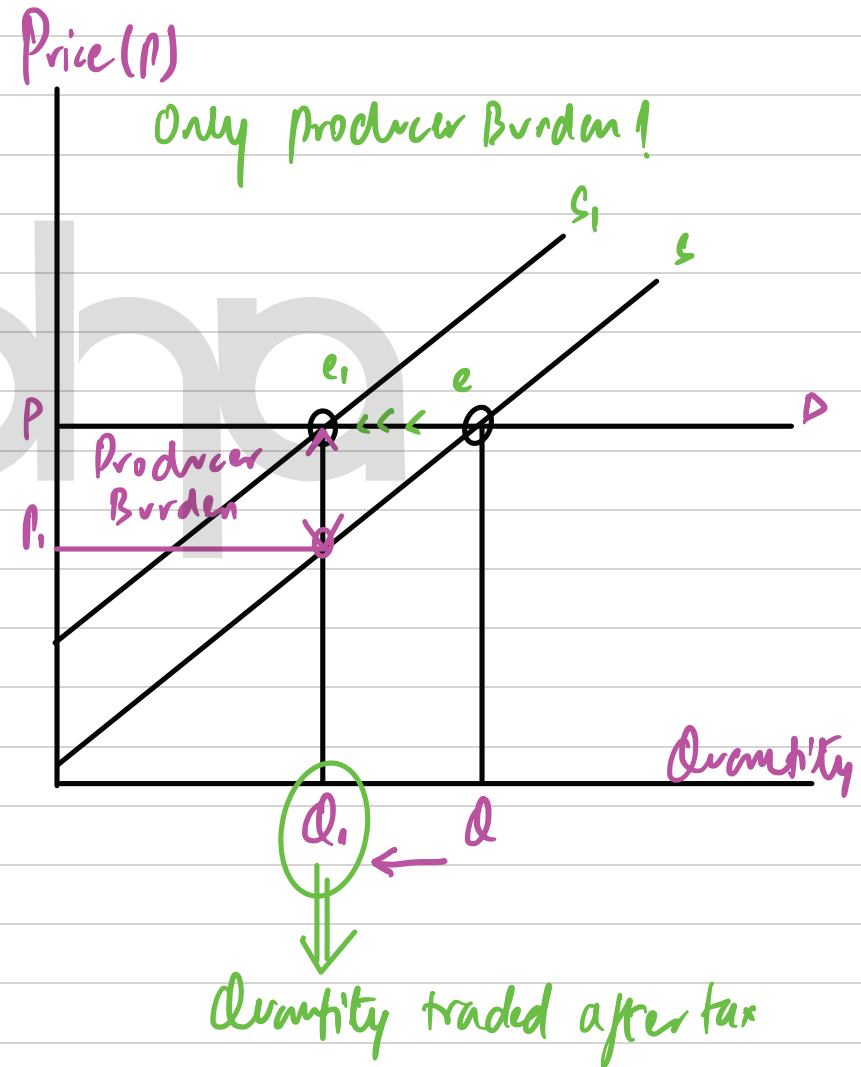
Objective:- Generating tax revenue by imposing tax on harmful goods (Demerit goods).

Quantity traded after tax!

Indirect tax on
perfectly inelastic
demand



Indirect tax on
perfectly elastic
demand!



Questions

Marks :- 16

Q1 Using a diagram, explain why a government would impose indirect tax on luxury good ? (8)

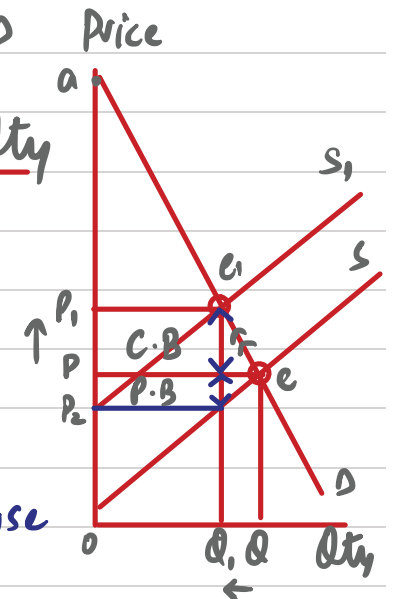
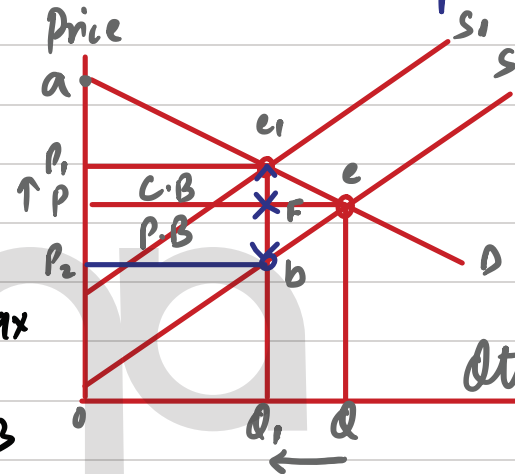
Ans :- Define indirect tax, e.g. (sales tax)

Diagram showing impact of indirect tax

↳ Impact on C.S, P.S, TR, C.B + P.B

Explain using example

1. Tax on imported cars
2. Tax on private transport



Q2 Using a diagram, explain why a government would impose indirect tax on harmful goods ? (8)

indirect tax on harmful goods ? (8)

Indirect tax with inelastic demand (Diagram)

e.g. Demerit goods
Impact on C.B and P.B !